Supplementary Material for FiHaRD

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1 Theoretical Properties of the Spatio-Spectral Forward Process

Proposition 1 (Validity of the spatio-spectral forward process). *Let*

$$\Sigma_{\text{space}} = U \Lambda U^\top, \qquad \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_F),$$

with U orthonormal and $\lambda_m \geq 0$, and let $\nu_\ell \geq 0$ denote the frequency weights associated with the future DCT modes. Define the joint mode energies

$$\kappa_{\ell,m} = \nu_\ell \lambda_m,$$

and assume that the cumulative retention factors satisfy $0 < \bar{\alpha}_{t,\ell,m} \leq 1$ for every timestep t and every spatio-spectral mode (ℓ, m). Then the forward process

$$z_t = \sqrt{\bar{\alpha}_t} \odot z_0 + \sqrt{1 - \bar{\alpha}_t} \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I),$$

with $z_0 = r_0 U$, is:

1. *admissible, since each mode has nonnegative variance $1 - \bar{\alpha}_{t,\ell,m}$;*
2. *linear-Gaussian in the residual coordinates, because it is diagonal Gaussian in z-space and mapped back by the orthonormal transform $r = zU^\top$;*
3. *Markov, with one-step coefficients $\alpha_{t,\ell,m} = \bar{\alpha}_{t,\ell,m} / \bar{\alpha}_{t-1,\ell,m}$ defined mode by mode; and*
4. *compatible with v-prediction, since together with z_t, the target*

$$v_t = \sqrt{\bar{\alpha}_t} \odot \epsilon - \sqrt{1 - \bar{\alpha}_t} \odot z_0$$

uniquely determines both z_0 and ϵ mode by mode.

2 Proof of Proposition 1

We prove each statement in turn.

Admissibility. By assumption, every cumulative retention factor satisfies

$$0 < \bar{\alpha}_{t,\ell,m} \leq 1.$$

Therefore each mode of the forward process has variance

$$1 - \bar{\alpha}_{t,\ell,m} \geq 0,$$

so the noising rule

$$z_{t,\ell,m} = \sqrt{\bar{\alpha}_{t,\ell,m}}\, z_{0,\ell,m} + \sqrt{1 - \bar{\alpha}_{t,\ell,m}}\, \epsilon_{\ell,m}$$

defines a valid Gaussian perturbation for every spatio-spectral mode.

Linear-Gaussian structure. In the spectral coordinates, the process is diagonal across modes and Gaussian because $\epsilon \sim \mathcal{N}(0, I)$ and the update is affine in (z_0, ϵ). The residual coordinates are obtained through the linear map

$$r_t = z_t U^\top,$$

with U orthonormal. Linear images of Gaussian variables remain Gaussian, hence the induced process in residual coordinates is also linear-Gaussian.

Markov property. Let

$$\alpha_{t,\ell,m} = \frac{\bar{\alpha}_{t,\ell,m}}{\bar{\alpha}_{t-1,\ell,m}}.$$

Then the cumulative coefficients factorize exactly as in standard diffusion, mode by mode. Consequently, each coordinate admits a one-step transition of the form

$$z_{t,\ell,m} = \sqrt{\alpha_{t,\ell,m}}\, z_{t-1,\ell,m} + \sqrt{1-\alpha_{t,\ell,m}}\, \eta_{t,\ell,m}, \qquad \eta_{t,\ell,m} \sim \mathcal{N}(0,1),$$

which depends on the past only through $z_{t-1,\ell,m}$. Since this holds independently for every mode, the full process is Markov.

Compatibility with v-prediction. For each mode, define

$$A_{t,\ell,m} = \sqrt{\bar{\alpha}_{t,\ell,m}}, \qquad B_{t,\ell,m} = \sqrt{1-\bar{\alpha}_{t,\ell,m}}.$$

The forward process and the v-target are

$$z_{t,\ell,m} = A_{t,\ell,m} z_{0,\ell,m} + B_{t,\ell,m} \epsilon_{\ell,m},$$

$$v_{t,\ell,m} = A_{t,\ell,m} \epsilon_{\ell,m} - B_{t,\ell,m} z_{0,\ell,m}.$$

This is a 2×2 linear system in the unknowns $(z_{0,\ell,m}, \epsilon_{\ell,m})$. Its coefficient matrix

$$\begin{bmatrix} A_{t,\ell,m} & B_{t,\ell,m} \\ -B_{t,\ell,m} & A_{t,\ell,m} \end{bmatrix}$$

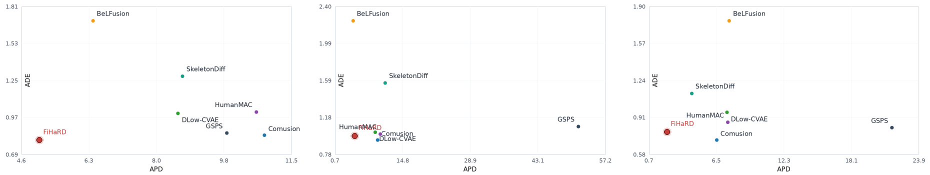
has determinant

$$A_{t,\ell,m}^2 + B_{t,\ell,m}^2 = \bar{\alpha}_{t,\ell,m} + (1-\bar{\alpha}_{t,\ell,m}) = 1.$$

Hence it is invertible, so $(z_{0,\ell,m}, \epsilon_{\ell,m})$ is uniquely determined by $(z_{t,\ell,m}, v_{t,\ell,m})$ for every mode. Therefore the proposed forward process is compatible with standard v-prediction.

3 Diversity-Quality Tradeoff

The quantitative results in the main paper suggest that FiHaRD occupies a favorable region of the diversity-quality tradeoff, especially when quality is evaluated jointly through forecasting accuracy and distributional fidelity. In our



(a) Assembly. Each point is averaged over the available action-specific runs for that model.

(b) H2O.

(c) BigHands.

Fig. 1: Diversity-quality tradeoff measured as APD versus ADE on the three benchmarks. Higher APD indicates greater diversity, while lower ADE indicates better forecast quality. Across the three datasets, the plots show that the largest diversity values do not automatically correspond to the best forecast quality, and that FiHaRD tends to remain in a favorable low-error region without requiring extreme APD.

tables, APD serves as a diversity indicator, while ADE/FDE, CMD, and FID reflect reconstruction quality and sample consistency with the target distribution.

The comparison shows that the methods with the largest APD are not necessarily the strongest overall generators. For example, on Assembly actions such as *attempt*, *remove*, and *unscrew*, gsps achieves the highest APD, but FiHaRD attains better ADE and substantially better FID. A similar pattern appears on BigHands, where gsps again provides the highest APD, yet FiHaRD remains clearly stronger in CMD and FID while also being more competitive in ADE. On H2O, FiHaRD does not maximize APD either, but it still maintains a better balance between sample quality and realism than several older baselines.

We interpret this behavior as consistent with the proposed spatio-spectral residual diffusion design. Because uncertainty is injected along articulated spatial modes and future-frequency modes, the model does not need to increase diversity through unconstrained variance alone. Instead, it can preserve relatively low distributional error while still generating multiple plausible futures. In this sense, the theoretical structure of the forward process appears to regularize the diversity-quality tradeoff: diversity is retained where the motion forecast is genuinely multimodal, but the samples remain closer to the geometry and statistics of realistic hand motion.